



Management Science

Publication details, including instructions for authors and subscription information:
<http://pubsonline.informs.org>

Reducing Capital Market Anomaly: The Role of Information Technology Using an Information Uncertainty Lens

Ning Jia, Arun Rai, Sean Xin Xu

To cite this article:

Ning Jia, Arun Rai, Sean Xin Xu (2020) Reducing Capital Market Anomaly: The Role of Information Technology Using an Information Uncertainty Lens. Management Science 66(2):979-1001. <https://doi.org/10.1287/mnsc.2018.3235>

Full terms and conditions of use: <https://pubsonline.informs.org/Publications/Librarians-Portal/PubsOnLine-Terms-and-Conditions>

This article may be used only for the purposes of research, teaching, and/or private study. Commercial use or systematic downloading (by robots or other automatic processes) is prohibited without explicit Publisher approval, unless otherwise noted. For more information, contact permissions@informs.org.

The Publisher does not warrant or guarantee the article's accuracy, completeness, merchantability, fitness for a particular purpose, or non-infringement. Descriptions of, or references to, products or publications, or inclusion of an advertisement in this article, neither constitutes nor implies a guarantee, endorsement, or support of claims made of that product, publication, or service.

Copyright © 2019, INFORMS

Please scroll down for article—it is on subsequent pages



With 12,500 members from nearly 90 countries, INFORMS is the largest international association of operations research (O.R.) and analytics professionals and students. INFORMS provides unique networking and learning opportunities for individual professionals, and organizations of all types and sizes, to better understand and use O.R. and analytics tools and methods to transform strategic visions and achieve better outcomes. For more information on INFORMS, its publications, membership, or meetings visit <http://www.informs.org>

Reducing Capital Market Anomaly: The Role of Information Technology Using an Information Uncertainty Lens

Ning Jia,^a Arun Rai,^b Sean Xin Xu^c

^a School of Economics and Management, Tsinghua University, 100084 Beijing, People's Republic of China; ^b Center for Process Innovation and Department of Computer Information Systems, Robinson College of Business, Georgia State University, Atlanta, Georgia 30303; ^c Research Center for Contemporary Management, Key Research Institute of Humanities and Social Sciences at Universities, School of Economics and Management, Tsinghua University, 100084 Beijing, People's Republic of China

Contact: jian@sem.tsinghua.edu.cn (NJ); arunrai@gsu.edu (AR); xuxin@sem.tsinghua.edu.cn,  <http://orcid.org/0000-0002-9453-8404> (SXX)

Received: February 10, 2017

Revised: April 22, 2018; September 18, 2018

Accepted: October 2, 2018

Published Online in Articles in Advance:
August 5, 2019

<https://doi.org/10.1287/mnsc.2018.3235>

Copyright: © 2019 INFORMS

Abstract. We investigate how firms use information technology (IT) implementation to mitigate an anomaly in capital markets: investors underreacting to new public information. The theory of information uncertainty (IU) suggests that the anomaly is amplified with IU; that is, with ambiguity in information about firm value. We theorize that a firm's IT in general—and enterprise systems (ES) in particular—can mitigate IU, thus reducing the IU-induced underreaction anomaly. Based on a difference-in-differences analysis of a sample of 572 ES implementations, our main finding is that ES implementation does reduce IU-induced underreaction anomaly. This is achieved through a reduction in the firm's fundamentals volatility and an improvement in information quality. We also find that firms with greater IT capability are better positioned to realize the anomaly-reducing benefits of ES implementation and that ES's anomaly-reducing effect is most pronounced when high levels of both functional and operational ES modules are implemented. We obtain remarkably consistent results when using alternate empirical design, samples, and measures of news. Such IT impacts are economically highly consequential because they improve capital market efficiency.

History: Accepted by Anandhi Bharadwaj, information systems.

Funding: N. Jia acknowledges the financial support of the National Natural Science Foundation of China [Grant 71372049] and the BNP Paribas-Tsinghua SEM Center for Globalization of Chinese Enterprises. S. X. Xu acknowledges financial support from the National Natural Science Foundation of China [Grant 71490724].

Supplemental Material: The data files and online appendices are available at <https://doi.org/10.1287/mnsc.2018.3235>.

Keywords: capital market • anomaly • market efficiency • information uncertainty • enterprise systems • IT business value

1. Introduction

We examine the impact of information technology (IT) implementations by public companies on a capital market anomaly. We shed light on “market efficiency” (Fama 1965) and contribute to the literature on IT business value by identifying a market-level impact of firm IT, whereas most studies focus on more micro levels, such as firm productivity and profitability (Kohli and Grover 2008).

Our study has economically significant implications. The capital market allocates ownership of the economy's capital stock. Ideally, it aggregates information about each firm's economic activities in a timely and comprehensive manner to help market participants choose securities and level of ownership. In an efficient market, new information alters investors' valuation and thus triggers changes in stock prices, so that they can be accurate signals for capital allocation (Fama 1965). Market efficiency is also a major regulatory objective. The U.S. Securities

Exchange Act of 1934 states that “[i]n developing rules, the rules must be germane to the goal of market efficiency” and, more recently, the U.S. Supreme Court stated that “[t]he magnitude of the federal interest in protecting the integrity and efficiency of the national securities market cannot be overstated” (Huber 2016, p. 405).

A large body of research in accounting, finance, and economics, however, has produced mounting evidence across global markets that contradict Fama's market efficiency hypothesis (for a review, see Daniel et al. 1998). These phenomena are collectively known as capital market anomalies, which include both investor overreaction (Bondt and Thaler 1985) and underreaction (Jegadeesh and Titman 1993) to new information. They suggest that investors overreact/underreact to new information about firm value, and as a result, the price change associated with the new information is too large/small. These anomalies affect the efficient functioning of the capital market, because security mispricing distorts

the allocation of wealth between firms and investors (Kothari 2001). Any mechanism that can correct such anomalies therefore deserves systematic research.

For both practical and theoretical reasons, we focus on the potential of public firms' IT to mitigate the underreaction anomaly. This anomaly adversely impacts the capital market and listed firms. It delays the pricing of new information into stock prices, with significant economic consequences (Foster et al. 1984). Stock underpricing results not only in lower investment and employment (Hau and Lai 2013) but also in higher costs of capital, because investors require extra return to compensate for disproportionate stock moves (Lambert et al. 2007). Executives of undervalued firms are likely to receive lower stock option compensation and lower total compensation, harming managerial incentives (Forsyth et al. 2007). Systematic research on how to alleviate the underreaction anomaly is thus important for constituencies including policymakers and top management.

Prior studies have provided a solid theory—information uncertainty (IU) theory—that guides our theoretical development for the impact of firm IT on the underreaction anomaly. *Information uncertainty*—a driver of the underreaction anomaly (Jiang et al. 2005, Zhang 2006)—is defined as

ambiguity with respect to the implications of new information for a firm's value, which potentially stems from two sources: the volatility of a firm's underlying fundamentals and poor information. (Zhang 2006, p. 105)

If market efficiency holds, investors should react to new information in the market "instantaneously" and "completely" (Fama 1965). IU, however, induces underreaction (Jiang et al. 2005, Zhang 2006).¹ We consider the impact of a firm's IT on its IU and thus on the IU-induced underreaction anomaly.²

Our analysis unfolds as follows: In Section 2, we theorize how IT in general and enterprise systems (ES) in particular can mitigate the IU-induced underreaction anomaly. We choose ES as the focal IT for two reasons. First, it has been the largest category of IT investment since the 2000s and remains so, according to recent industry estimates (Reisinger 2014). ES is therefore representative of a firm's IT, at least in terms of the amount of capital investment, and thus its impact needs to be examined. Second, as suggested by prior research, ES can influence both a firm's information environment (e.g., Dorantes et al. 2013) and its fundamentals (e.g., Tian and Xu 2015), the driving forces for the IU-induced underreaction anomaly (Zhang 2006). We thus have a strong theoretical basis for focusing on ES and for formulating the main hypothesis that *ES implementation reduces the IU-induced underreaction anomaly*.

ES is not the only factor that affects information uncertainty or anomalies in the market³ and it can only help combat some aspects of IU. We expect ES implementation to mitigate IU-induced anomaly but not eliminate it. How significant is the ES impact? This is an interesting question deserving empirical research. Section 3 shows our main analysis. We hand-collected data on ES-implementation announcements, which allows us to use a difference-in-differences (DiD) research design to test ES's impact (Dorantes et al. 2013). We start by replicating Zhang's (2006) model to quantify the IU-induced underreaction anomaly. Then, our DiD tests demonstrate that firms implementing ES incur a significantly lower anomaly than matched firms do. We further investigate contingencies in the ES impact. We find that firms with greater IT capability are better positioned to realize the anomaly-reducing benefits of ES implementation and that ES's anomaly-reducing effect is most pronounced with high implementation levels of both functional and operational modules.

Section 4 documents supplementary and robustness tests, including a test of mediation models (e.g., ES reduces a firm's fundamentals volatility and improves information quality, which in turn reduces underreaction anomaly), an alternative matching algorithm for the DiD analysis, alternative criteria for data exclusion, and alternate measures of news. These findings support our main findings and shed light on the mechanisms by which ES mitigates the IU-induced underreaction anomaly.

Section 5 discusses our study's implications for research and for practice, its strengths and limitations, and directions for future research. Section 6 concludes.

2. Theoretical Development

2.1. The Market Underreaction Anomaly

It is well recognized that the formation of stock prices is affected by the speed and efficacy with which investors incorporate new information into their expectations about a firm's value. In an efficient capital market, prices adjust "instantaneously" and "completely" to new information (or "news") to reflect investors' changing expectations about a firm's value, so that at any point in time, an investor can only expect a normal rate of return on an investment in any stock (Fama 1965). However, a large body of accounting, finance, and economics research suggests otherwise. One well-documented empirical regularity is investor underreaction to new public information. That is, rather than correctly appreciating the implications of current news and immediately revising their expectations of firm value, investors recognize the full impact of news only *gradually* (see a seminal work by Ball and Brown 1968). The literature has repeatedly identified supportive evidence (e.g., Freeman and Tse 1989, Bernard and Thomas 1990, Jegadeesh and Titman 1993).

There are two behavioral explanations for investor underreaction to public news. The idea of *anchoring bias* (Tversky and Kahneman 1974) is that investors' estimates are often influenced by an initial starting value—their priors. Investors tend to overweight their priors, placing inadequate weight on public news (Campbell and Sharpe 2009). Another explanation is *investor overconfidence* (Daniel et al. 1998). Because in general an investor is more confident about signals or assessments with which (s)he has greater personal involvement, (s)he will be overconfident about the information (s)he has generated but not about public signals. Both behavioral biases lead to investor underreaction to new public information.

2.2. Information Uncertainty

Building upon this line of work, IU theory contends that the underreaction anomaly is amplified in cases of greater IU (Jiang et al. 2005, Zhang 2006), where IU is defined in terms of “value ambiguity,” or the degree to which a firm's value can be reasonably estimated by even the most knowledgeable investors (Daniel et al. 1998, Hirshleifer 2001, Jiang et al. 2005). IU theory states that, although IU is latent, it is manifested in the underreaction anomaly (Zhang 2006). That is, with greater value ambiguity, investors are even more likely to overweight their priors or private signals and underreact to public information, leading to a more severe underreaction anomaly.

IU may stem from two sources: the volatility of a firm's underlying fundamentals, and poor information. As formalized in Zhang (2006), an observed signal (s) (that is, new information) is characterized as a firm's fundamental value (v) plus a noise term (e); that is, $s = v + e$. Assuming the relationship between noise and fundamental value is orthogonal— $\text{cov}(v, e) = 0$ —then the variance of the signal measures IU; that is, $\text{var}(s) = \text{var}(v) + \text{var}(e)$, where $\text{var}(v)$ is a firm's underlying fundamentals volatility, and $\text{var}(e)$ reflects the quality of information. We use the two notations “ $\text{var}(v)$ ” and “ $\text{var}(e)$ ” throughout the rest of the paper. High volatility of fundamentals—high $\text{var}(v)$ —implies that the firm's expected future cash flows are less “certain” or “knowable” and are inherently difficult to predict, making estimates of its fundamental value less reliable. Poor information—high $\text{var}(e)$ —means that the information that the firm supplies to the market is an inaccurate representation of its economic reality, which reduces the usefulness of the signal in inferring the firm's value.

2.3. ES Mitigation of the IU-Induced Underreaction Anomaly

The main idea we test is that by implementing ES, firms can mitigate the IU-induced underreaction anomaly. We submit that *ES can help reduce IU so that*

information provided by firms is more certain and investors will be more responsive to it. We summarize the theoretical rationales underlying the mitigation effect of ES implementation in Table 1 and elaborate them in detail below.

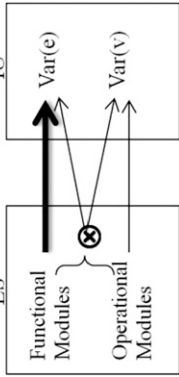
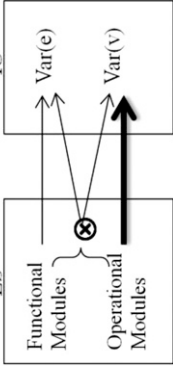
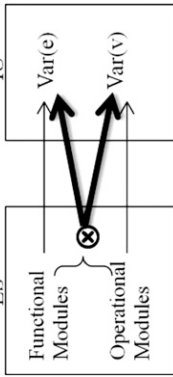
There are two types of ES module. *Functional modules* support the administrative processes of a firm's value chain, such as accounting, whereas *operational modules* support the operational processes of a firm's value chain, such as sales and procurement (Barki and Pinsonneault 2005, Ranganathan and Brown 2006). Major functional modules include accounting, finance, and human resource management. Accounting and finance modules are closely related to IU because they produce financial reports that convey a firm's information to investors. Therefore, in the rest of the paper, when we talk about ES functional modules, we are referring to accounting and finance modules. Major *operational modules* include (a) customer and sales modules and (b) supply chain modules, which subsume internal operations and supplier-related operations.

Table 1 presents the pathways by which ES modules influence IU (“Graphical illustration” column). Three mechanisms underlie the pathways: *automation*, *standardization*, and *integration*. Through these, functional modules improve the information investors receive from the firm, thereby reducing $\text{var}(e)$, whereas operational modules enable firms to respond in a more coordinated and prompt manner to changes in the external environment, thereby reducing performance volatility $\text{var}(v)$. The interaction between functional and operational modules further reduces $\text{var}(e)$ and $\text{var}(v)$.

2.3.1. Automation. ES can automate functional and operational processes of a firm's value chain. The automation feature of functional modules enables automated processing of large amounts of data and the production of timely, accurate information about the firm's economic activities and performance. Automated processes are less prone to human error and produce more-accurate data. Automated financial transactions can reduce cash-to-cash cycle times and the time needed to reconcile financial data at quarter- or year-end (Stratman 2001, Mabert et al. 2003), so the information is more up to date and better reflects the underlying economic reality. Both accuracy and timeliness increase information quality (Wixom and Todd 2005), reducing $\text{var}(e)$.

The automation feature of operational modules can reduce operational volatility $\text{var}(v)$. When the operating environment changes, ES can automatically adjust value chain processes (Gattiker and Goodhue 2005, Tian and Xu 2015). It can eliminate manual work and reduce the need to resolve the data incompatibilities common to legacy systems. Thus, a firm can make timely

Table 1. Underlying Rationales for ES Mitigation of Information Uncertainty

	Automation	Standardization	Integration	Graphical illustration
ES functional modules (accounting and finance)	Automated information processing enhances information timeliness and accuracy, thereby reducing var(e) . For example, automated financial transactions reduce cash-to-cash cycle times and the time needed to reconcile financial data at the end of the quarter or year, so that financial information reflects firm performance in a timely and accurate manner.	Standardized accounting processes improve information quality by ensuring that accounting data are collected using consistent and reliable processes and standard chart of accounts, thereby reducing var(e) . For example, sales personnel in different regions consistently follow the same process-embedded protocols when creating new sales records.	Integration of functional processes guarantees consistent data formats, regulates data compatibility and completeness, and detects data transmission errors, thereby reducing var(e) . For example, when processing an invoice, ES performs a three-way matching of the amounts on the invoice against the amounts on the receipt and the initial purchase order, reducing human error and increasing data accuracy (Davenport 1998).	
ES operational modules (customer and sales, supply chains)	ES operational modules enable automated reaction to changes in operating environments, thus reducing var(v) . For example, when customers change their orders, ES automatically triggers responses in customer and sales management and in supply chain management, thus reducing performance variability due to environmental changes.	By imposing predefined operational processes, ES operational modules reduce employees' and managers' discretion, thus reducing var(v) . For example, a manufacturer uses ES to standardize operational processes for order specification, configuration, and quality assurance, reducing variance in the execution of these processes and smoothing performance over time (Cotteleur and Bendoly 2006).	Integration of operational processes enables interdependencies between processes to be coordinated responsively, allowing the value chain to work as a whole to sense and respond to environmental changes, thereby reducing var(v) . For example, connected by enterprise-wide ES, a firm's distribution centers can coordinate to ship goods to cope with local demand shocks, mitigating the impact of such shocks on the firm's overall operating performance.	
Complementarity between functional and operational modules	Operational modules enhance the automatic functioning of accounting modules, further enhancing the accuracy and currency of accounting information and reducing var(e) . For example, PepsiAmericas implemented ES operational modules for inventory management. Prior to that, portions of the company's monthly inventory accounting had to be done manually and took two weeks. ES operational modules fed accounting modules with needed data, allowing inventory accounting to be done at period end automatically (PeopleSoft 2005).	The standardization effect of operational modules can be stronger when coupled with functional modules, further reducing var(v) . For example, selling requires support from accounting processes such as checking product codes, verifying customer credit, and settling accounts and transactions. To the extent that functional modules standardize those processes, they work together with operational modules to standardize how a firm conducts its operations. Greater benefits of standardization can be expected.	With ES functional modules, managers make informed decisions to cope with uncertainties; with ES operational modules, the decisions can be executed efficiently along enterprise value chains. Functional and operational modules thus work together to mitigate operational volatility, further reducing var(v) . For example, facing supply disruptions, functional modules fed Dell management with data on sales, parts and inventory, and production costs for each product line, so that management could make informed decisions on how to adjust product prices. Operational modules allowed Dell to change product prices and reschedule production across product lines efficiently (Lee 2004).	

Notes. An observed signal (s) for firm value assessment is characterized as a firm's fundamental value (v) plus a noise term (e); that is, $s = v + e$. The variance of the signal measures $IU: \text{var}(s) = \text{var}(v) + \text{var}(e)$, where $\text{var}(v)$ is volatility of firm fundamentals, and $\text{var}(e)$ reflects the quality of information. Table entries state how ES modules mitigate $\text{var}(e)$ and $\text{var}(v)$.

and efficient adjustments to changes in its operating environment. For instance, when customers change their orders, ES automatically triggers responses in customer and sales management (for example, updating customer purchases, signaling opportunities for cross-sales and upsales, and scheduling fulfillment and support) and in supply chain management (for example, changing production plans, monitoring inventory, adjusting work schedules, sending purchase orders to suppliers, and coscheduling with business partners) (Tian and Xu 2015). When there are problems with internal production processes, such as bottlenecks and stockouts, ES updates downstream and upstream processes in real time (Park and Kusiak 2005). By enabling a firm to respond efficiently and in a timely and coordinated manner to changes in its operating environment, ES can reduce performance volatility, thus $\text{var}(v)$.

2.3.2. Standardization. ES standardizes a firm's business processes (Cotteleer and Bendoly 2006). Standard functional processes enhance information quality by ensuring that accounting data are collected using a standard chart of accounts (Brazel and Dang 2008). For instance, sales personnel in different regions must follow the same process-embedded protocols when creating new sales records. Such approaches achieve compliance with proper accounting practices and standards, which in turn ensures comparability of sales numbers from different locations. Hence, standard accounting processes embedded in ES functional modules increase accounting data quality through the execution of consistent and repeatable processes, thus reducing $\text{var}(e)$.

Standardizing operational processes means that they all follow predetermined protocols and can be better controlled (Cotteleer and Bendoly 2006). The design of such system-defined protocols with configurable parameters is typically based on industry best practices. ES guides all employees to comply with the same rules for decision and action, thus reducing variation in operational outcomes due to idiosyncratic personal discretion (Brazel and Agoglia 2007). Cotteleer and Bendoly (2006, p. 645) document a manufacturer's use of ES to standardize order specification, configuration, and quality assurance, reducing variance in the execution of these processes and smoothing performance over time; that is, reducing $\text{var}(v)$.

2.3.3. Integration. In addition to automation and standardization, integration of business processes is a key advantage of ES (Davenport 1998, Weill and Ross 2009). Integration means that business processes are interdependent and responsive to each other (Barki and Pinsonneault 2005). The integration is realized through the use of a central database, where business processes extract data needed for operations and save new data generated by operations (Davenport 1998).

The integration of functional modules guarantees consistent formats for data exchanged between accounting and finance processes, regulates compatibility and completeness of transmitted data, and helps detect data transmission errors (Davenport 1998, Barki and Pinsonneault 2005). For instance, when headquarters aggregate regional accounting data, an enterprise-wide ES retrieves data saved by the accounting processes of the headquarters and the branches. The central database defines formats and regulates how to aggregate data from separate processes, eliminating data errors that may occur across otherwise separate processes (Markus et al. 2000). Integration also facilitates cross-validation of information across processes. For example, when an accounts payable clerk enters a vendor's invoice into the system, it is stored in the database. When the vendor's merchandise is received, a receipt is created and also entered. Before the accounts payable department pays the vendor, the ES accounting module compares the amounts on the invoice, the receipt, and the purchase order. If they match, payment is approved (Davenport 1998). In this way, integrated accounting and finance processes reduce human error across processes. With improved accounting data, management can increase the quality of information that describes firm fundamentals, thereby reducing $\text{var}(e)$.

Integrating operational modules helps operational processes along the entire value chain work as a whole to cope with environmental changes (Weill and Ross 2009, Tian and Xu 2015). For instance, enterprise-wide ES can coordinate shipment of goods between distribution centers to be responsive in real time to local demand shocks, mitigating adverse effects on overall operating performance (Lee 2004). As another example, consider the so-called "bullwhip effect," in which delays in transmitting information about demand changes to upstream supply chain stages cause stockouts or excess echelon inventory, increasing the variability of sales and inventory costs (Lee et al. 1997). ES operational modules, by enhancing coordination between downstream customer-facing processes and upstream supply chain processes, enable the firm to sense and respond to demand changes in a more integrated and timely manner (Yao and Zhu 2012), reducing operational volatility $\text{var}(v)$.

2.4. Complementarity of ES Functional and Operational Modules

In addition to individually reducing the IU-induced underreaction anomaly (as theorized above), we expect ES functional modules (accounting and finance) and operational modules (customer and sales, supply chain) to have a complementary effect that further reduces the anomaly.

First, operational modules strengthen the automation feature of functional modules. Every operational process involves information-processing tasks (Porter and Millar 1985); ES operational modules digitize operational processes pertaining to customers and sales as well as internal operations and supplier processes, storing the needed data in centralized databases with defined formats. ES functional modules can retrieve the operational data (Davenport 1998), further automating and accelerating the accounting and finance processes. For example, PepsiAmericas implemented ES operational modules for inventory management. Some monthly inventory accounting had been done manually, which took two weeks. Now, ES operational modules feed the necessary data to accounting modules, so that inventory accounting is automatically done at period end (PeopleSoft 2005). As another example, ES operational modules collect and disseminate timely operational data (such as customer relations) to managers, improving their ability to process and analyze related accounting information/accruals (such as allowance for doubtful accounts). Managers can provide financial reports more frequently and with more accurate estimates of the firm's earnings (Li et al. 2012). In sum, operational modules enhance the automatic functioning of the functional modules in a consistent manner, improving the accuracy and timeliness of accounting information and reducing $\text{var}(e)$.

The second underlying rationale for complementarity between functional and operational modules is that the standardization effect of operational modules can be stronger when coupled with functional modules. Functional processes are also called "supportive" processes because they support operational processes (Porter and Millar 1985). Selling, for example, calls for the support of such accounting processes as checking product codes, verifying customer credit, and settling accounts and transactions. To the extent that functional modules standardize those accounting processes, they work with operational modules to standardize selling operations. Greater benefits of standardization can thus be expected, such as reducing human error and variation due to personal discretion. For example, a manufacturer using ES operational modules to manage procurement will face a delay in replenishment if a procurement manager enters incorrect product codes. Such human error can be avoided if ES accounting modules standardize information (products, accounts, and so on), save it in centralized enterprise-wide repositories, and update it routinely (Mukhopadhyay and Kekre 2002). Operational and functional modules would work together to standardize the procurement process and ensure that operational data are error-free, eliminating unexpected delays and, more broadly, reducing $\text{var}(v)$.

Third, by integrating a firm's accounting processes, ES functional modules act as an information hub, aggregating real-time information across product lines, business divisions, and regions. A firm's accounting system is supposed to generate data to describe its economic activities; with ES functional modules, managers have more timely and accurate information with which to make better decisions (Lambert et al. 2007). In particular, they can better sense and respond to market uncertainties (Sambamurthy et al. 2003), reducing $\text{var}(v)$. When an earthquake caused supply disruptions in Taiwan in the late 1990s, most personal computer manufacturers were unable to deliver on time and experienced high fluctuations in performance (Lee 2004). But Dell, with an ES that provided data on sales, parts, inventory, and production costs for each product line, could adjust prices across its product lines overnight, steering demand away from products built with parts that were unavailable because of the earthquake and toward others that did not use those parts (Lee 2004). In this case, Dell relied on two pillars of ES. One was functional modules to represent economic realities across regions and product lines, so that top management could make informed decisions. The other was operational modules that standardized and integrated operational processes, allowing Dell to change prices and reschedule production across product lines efficiently. In short, ES functional modules help managers make informed decisions to cope with uncertainties, whereas ES operational modules help them execute those decisions efficiently along enterprise value chains, collectively reducing $\text{var}(v)$.

In sum, ES functional and operational modules, individually and collectively, reduce IU through automation, standardization, and integration (summarized in Table 1). To the extent that IU is reduced, we expect the IU-induced underreaction anomaly to be mitigated. We hypothesize:

Hypothesis 1. *ES implementation mitigates the IU-induced market underreaction anomaly.*

3. Main Test: ES Impact on the IU-Induced Underreaction Anomaly

3.1. Proxies for IU

Information uncertainty is latent. To operationalize our research, we follow Zhang (2006), who proposed six proxies for IU. These are the *ex ante* conditions for high IU; that is, they are measured *before* the observed IU-induced anomaly (Zhang 2006).

Firm size ($1/MV$) is measured as the reciprocal of market capitalization. Small firms, with less information available for the market than large firms, generally have higher $\text{var}(e)$ (Bhushan 1989). They may also have fewer customers, suppliers, and shareholders, thus

being subject to high volatility of fundamentals and high failure rate (Mitchell 1994). A larger value of $(1/MV)$ indicates higher IU.

Firm age ($1/AGE$) is measured as the reciprocal of the number of years since the firm first appeared in the Center for Research in Security Prices (CRSP) database. Firms with a long history have more information available to the market, which implies a smaller $\text{var}(e)$ (Barry and Brown 1985). A larger value of $1/AGE$ indicates higher IU.

Analyst coverage ($1/COV$) is measured as the reciprocal of the number of analysts following the firm in the previous year. Analyst forecast has been widely used to measure the quality of the information environment (Chan and Hameed 2006). To the extent that analyst research is at least partially independent and different analysts' forecasts and reports are not released at the same time, a greater number of analysts reporting about a firm should improve information flow to market participants, implying a lower IU. As with firm size and age, a larger $1/COV$ implies higher IU.

Dispersion in analyst forecasts ($DISP$) is measured as the standard deviation of analyst forecasts made four months before fiscal year-end, scaled by the prior year-end stock price to mitigate heteroskedasticity. In the accounting and finance literatures, analyst forecast dispersion is widely used to proxy for uncertainty or the degree of consensus among analysts or market participants (Lang and Lundholm 1996). A larger $DISP$ indicates higher IU.

Stock volatility ($SIGMA$) is measured as the standard deviation of weekly market excess returns over the past year. Stock volatility reflects market perception about how volatile a firm's fundamentals and operations are. A larger $SIGMA$ indicates higher IU.

Cash flow volatility ($CVOL$) is measured as the standard deviation of cash flow from the previous five years' operations. It reflects the volatility of a firm's fundamentals and operations; larger $CVOL$ indicates higher IU.

These proxies may differ in their abilities to capture the latent variable IU. To the extent that most, if not all, information in the market about a firm comes from the firm itself (Healy and Palepu 2001), proxies based on firm fundamentals ($1/MV$, $1/AGE$, and $CVOL$) may have a higher correlation with IU. Proxies based on analysts ($1/COV$ and $DISP$) may be confounded by analysts' own interpretations of firm-provided information, leading to a lower correlation with IU. $SIGMA$ is based on the firm's stock returns, which incorporate heterogeneous interpretations of firm information by investors, including noise traders (Brown 1999), so this is likely to be a noisy proxy. In the analysis below, we rely on the overall pattern in the results across the six proxies to draw conclusions.

3.2. Estimation of the IU-Induced Underreaction Anomaly

We follow Zhang's (2006) empirical method to estimate $ANOMALY_{Sample Window}^{Sample}$, the IU-induced underreaction anomaly for a sample of firms and for a given time window. It reflects *delayed* investor reaction to news for that sample during that window. The underreaction anomaly means that because investors underreact to public news, the price change associated with it is too small. Underreaction will be corrected gradually as investors recognize the full impact of news, but not "instantaneously," as assumed by the market efficiency hypothesis. An empirical manifestation of such information-processing bias is that a stock price drifts in the direction of news: good news continues to generate positive returns, and bad news continues to generate negative returns after disclosure (Daniel et al. 1998, Jiang et al. 2005, Zhang 2006).

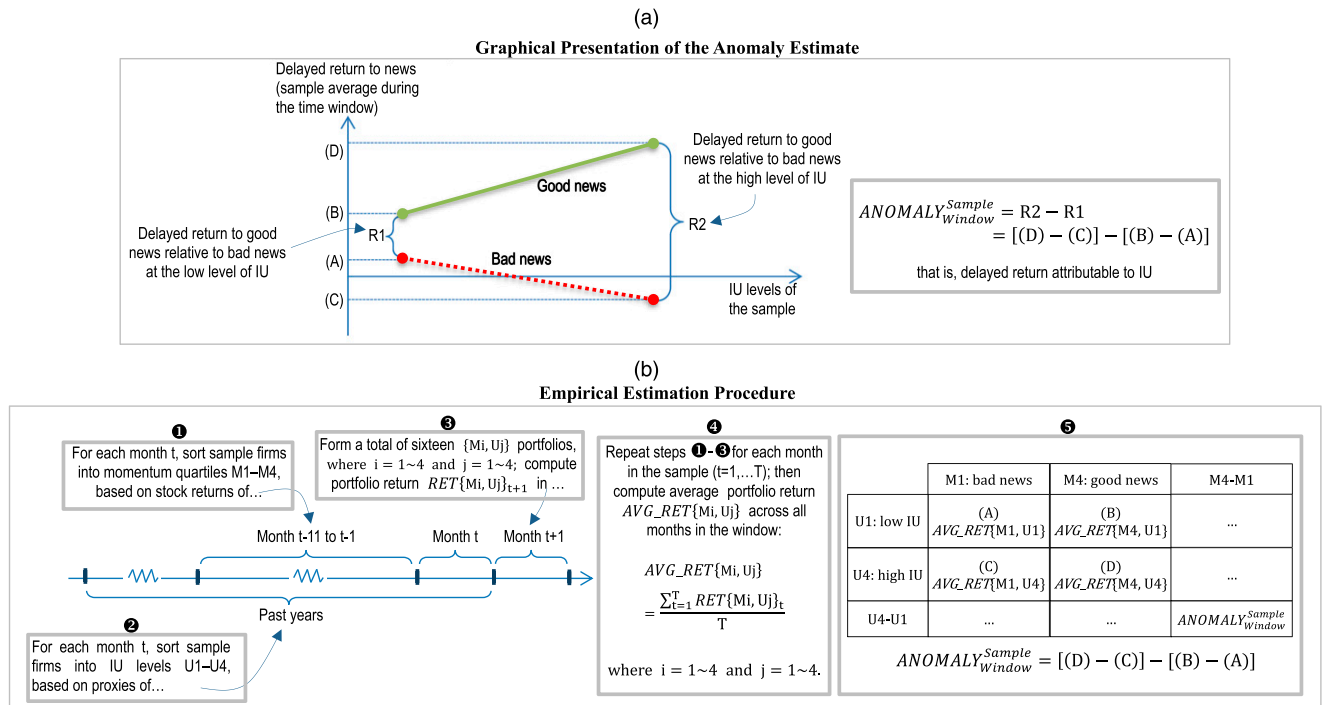
Our empirical estimation procedure assigns stocks into portfolios on the basis of good/bad news and levels of IU and examines the portfolios' average returns (Zhang 2006). A common measure of news is price momentum—cumulative stock returns over the previous 11 months. Stocks that performed well/poorly in the past imply good/bad news (Zhang 2006). Figure 1 illustrates how to empirically measure IU-induced underreaction anomaly, including five steps.

Step 1. For each month t in the window, we sort sample firms into price momentum quartiles, where the M1 (bad news) group contains firms with the lowest returns in the previous 11 months and the M4 (good news) group contains firm with the highest returns in the previous 11 months.

Step 2. For each momentum quartile, we sort sample firms—using each of the six IU proxies (Section 3.1 above)—into IU quartiles, U1 (lowest) through U4 (highest). To reiterate, the IU proxies are *ex ante*—based on data from past years. Thus, for each month in our sample period, on the basis of this double sorting (by momentum quartiles and then by IU quartiles), we classify sample firms into 16 groups/portfolios.

Step 3. For each month and each portfolio, we compute one-month-ahead portfolio returns, $RET\{Mi, Uj\}_{t+1}$, where $i = 1 \sim 4$ and $j = 1 \sim 4$. Consistent with Fama and French (1996) and Diether et al. (2002), we allow a one-month lag between the momentum measure and portfolio returns.

Step 4. We repeat Steps 1–3 for each sample month ($t = 1, \dots, T$), then compute average portfolio return, $AVG_RET\{Mi, Uj\}$, for each portfolio across all months. We then obtain a 4×4 matrix with different combinations of news and IU. The number reported in each cell is the average return of stocks that fall in a particular news–IU combination.

Figure 1. (Color online) Illustration of How to Empirically Measure IU-Induced Underreaction Anomaly

Notes. This figure illustrates how to empirically measure information uncertainty (IU)-induced underreaction anomaly for a sample of firms during a time window (labeled as $ANOMALY_{Sample_Window}$). As illustrated in panel (a), $ANOMALY$ refers to the delayed (stock market) return—attributed to increased IU—to good news relative to bad news. Panel (b) illustrates the empirical estimation procedure. Specifically, at step 1, we sort sample firms into four momentum quartiles M1–M4, where M1 indicates bad news and M4 indicates good news; at step 2, for each momentum quartile, we sort sample firms into four IU levels U1–U4, where U1 indicates lowest IU and U4 indicates highest IU; at step 3, we compute 1-month-ahead portfolio returns $RET\{M_i, U_j\}_{t+1}$, where $i = 1 \sim 4$ and $j = 1 \sim 4$; we repeat the above steps 1–3 for each month in the sample ($t = 1, \dots, T$) and at step 4, compute average portfolio return $AVG_RET\{M_i, U_j\}$ across all months in the window; lastly, step 5 gives the estimated $ANOMALY_{Sample_Window}$, which refers to the sample average, during the window, of the delayed return to good news relative to bad news at the high level of IU—that is, $AVG_RET\{M4, U4\} - AVG_RET\{M1, U4\}$, or, (D)-(C) as labeled in (a)—minus the delayed return to good news relative to bad news at the low level of IU—that is, $AVG_RET\{M4, U1\} - AVG_RET\{M1, U1\}$, or, (B)-(A) as labeled in (a).

Step 5. We calculate the estimated IU-induced underreaction anomaly, $ANOMALY_{Sample_Window}$, which refers to the sample average, during the window, of the delayed return to good news relative to bad news at the high level of IU ($AVG_RET\{M4, U4\} - AVG_RET\{M1, U4\}$) minus the delayed return to good news relative to bad news at the low level of IU ($AVG_RET\{M4, U1\} - AVG_RET\{M1, U1\}$).

In Step 5 it is important to focus on M4-M1—that is, (B)-(A) and (D)-(C)—because M4-M1 resembles a momentum trading strategy for each IU quartile, which means taking a long position in past winners (good news firms) and a short position in past losers (bad news firms). Momentum strategies are commonly used in finance research (see, for example, Zhang 2006) to help gauge the economic significance of results. A further minus operation between them [(D)-(C)] - [(B)-(A)], which takes the difference in return between the high-IU group and the low-IU group, helps isolate IU impact. A positive value suggests that momentum return increases with IU; that is, as shown in panel (a) of Figure 1, investors underreact more (greater delayed return) with greater information uncertainty. These momentum/

arbitrage returns represent the degree of market inefficiency (that is, anomaly). The higher the arbitrage returns, the higher the degree of market inefficiency.

3.3. Data and Sample

We obtained return data from CRSP, analyst forecast data from the Institutional Brokers' Estimate System Detail File, and financial data from Compustat. Following prior research, we trimmed observations that could be erroneous.⁴

For ES, we used two data sources. First, we hand-collected ES implementation announcements from the Lexis-Nexis database for 1997–2014. We followed prior studies (Dehning et al. 2007, Hendricks et al. 2007) in using the search keywords for ES and obtained a sample of 748 ES implementation announcements (more details provided in Appendix A, available online). Using ES implementation announcements, we can conduct DiD tests to identify ES impact relative to a control sample (Dorantes et al. 2013). Second, we used a sample of firms for 2001–2008 from the Computer Intelligence (CI) database, another widely used data source for ES research (Dewan and Ren 2011, Tian and

Xu 2015). It provides annual data about whether a firm had implemented ES and, if so, the specific modules (accounting and finance, customer and sales, and supply chain). However, it does not give us the go-live date, so we cannot conduct DiD tests with these data. We can, however, evaluate the association of the IU-induced anomaly with module implementation.

Panel A of Table 2 defines the variables, and panel B presents descriptive statistics for our three samples. The first sample, which includes all firms covered by Compustat during our sample period (1997–2014), is used to verify that the IU-induced underreaction anomaly documented by prior studies (e.g., Zhang 2006) still exists for our sample period. The second sample, the subset of firms with ES announcements, is used to examine the anomaly-reducing effect of ES. The third sample, firms covered by the CI database between 2001 and 2008, is used to examine the complementarity effect between ES functional and operational modules.

3.4. Base Model: Existence of the IU-Induced Underreaction Anomaly

Our empirical estimation starts with verifying the existence of the IU-induced anomaly during our sample period, following the empirical approach described in Section 3.2. Results, reported in Table 3, are consistent

with those of Zhang (2006). For example, when using $1/AGE$ as a proxy for IU, firms in the lowest IU quartile (U1) and lowest momentum quartile (M1, bad news) earn an average 0.82% monthly return, whereas firms in the lowest IU quartile (U1) and highest momentum quartile (M4, good news) earn an average 1.04%. The return differential between these two quartiles is 0.22% (t -statistic = 0.59), reported in the “M4-M1” column in Table 3. On the other hand, in the highest IU settings (U4), firms in the lowest momentum quartile (M1, bad news) earn an average 0.04% monthly return, whereas those in the highest quartile (M4) earn an average 2.22%, a differential of 2.18% (t -statistic = 3.83). The IU-induced underreaction anomaly, $ANOMALY_{1997-2014}^{Compustat}$, is 1.96% ($R^2 - R^1 = 2.18\% - 0.22\%$), with a t -statistic of 5.20. The other five IU proxies also produce significant positive $ANOMALY_{1997-2014}^{Compustat}$. Together, the six proxies draw a consistent picture that the IU-induced underreaction anomaly exists during our sample period, which serves as a basis of our subsequent analysis.

3.5. Hypothesis Testing: ES Impact on IU-Induced Underreaction Anomaly

To test our hypothesis of an anomaly-reducing effect of ES, we conduct a DiD analysis that examines

Table 2. Definitions of Variables and Descriptive Statistics

Panel A. Definitions of variables									
Variable	Definition								
RET_{t+1}	Returns from month $t + 1$								
$RET_{t-11,t-1}$	Accumulated returns from months $t - 11$ to $t - 1$								
MV_t	Market capitalization (in millions of dollars) at the end of month t								
AGE_t	Number of years since the firm was first covered by CRSP by the end of year before month t								
COV_t	Number of analysts following the firm in the previous year								
$DISP_t$	Standard deviation of analyst forecasts in month t scaled by the prior year-end stock price								
$SIGMA_t$	Standard deviation of weekly market excess returns over the past 12 months ending month t								
$CVOL_t$	Standard deviation of cash flow from operations in the past five years (with a minimum of three years), where cash flow from operations is earnings before extraordinary items minus total accruals estimated from the balance sheet approach, scaled by average total assets								
Panel B. Descriptive statistics									
	Sample: All Compustat firms (1997–2014), $n = 478,999$ (firm-month)			Sample: ES announcements (1997–2014), $n = 61,505$ (firm-month)			Sample: CI database (2001–2008), $n = 117,160$ (firm-month)		
Variable	Mean	SD	Median	Mean	SD	Median	Mean	SD	Median
RET_{t+1} (%)	1.111	13.441	0.866	1.266	12.924	1.042	1.191	12.432	0.891
$RET_{t-11,t-1}$ (%)	17.046	71.049	8.000	15.275	52.902	9.650	19.181	53.889	10.942
MV_t	3,480.898	9,169.338	644.996	4,143.820	6,346.173	931.383	3,294.056	5,479.141	958.358
AGE_t	17.742	16.664	12	21.523	18.736	15	19.717	17.890	13
COV_t	11	9	8	13	10	11	11	9	9
$DISP_t$ (%)	0.643	1.299	0.205	0.450	0.969	0.142	0.392	1.182	0.136
$SIGMA_t$ (%)	5.966	3.448	5.191	4.960	2.685	4.315	5.668	3.089	4.944
$CVOL_t$	0.086	0.087	0.058	0.063	0.071	0.046	0.078	0.077	0.054

Note. SD, standard deviation.

Table 3. Baseline Results: The IU-Induced Underreaction Anomaly

	Momentum quartile				
	M1 (bad news)	M2	M3	M4 (good news)	M4 – M1
Panel A. Information uncertainty proxied by 1/MV					
U1 (low) (%)	0.83	0.92	0.95	1.17	0.34 (0.72)
U2 (%)	0.67	0.88	1.04	1.12	0.46 (0.94)
U3 (%)	0.40	0.88	1.05	1.24	0.83 (1.96)
U4 (high) (%)	0.04	0.79	1.16	1.54	1.51 (2.53)
U4 – U1 (%)	–0.79 (–1.85)	–0.13 (–0.31)	0.22 (0.57)	0.37 (0.83)	ANOMALY^{Compustat}_{1997–2014} =1.16*** (3.79)
Panel B. Information uncertainty proxied by 1/AGE					
U1 (low) (%)	0.82	1.00	1.04	1.04	0.22 (0.59)
U2 (%)	0.52	0.86	1.04	1.20	0.69 (1.75)
U3 (%)	0.37	0.76	1.03	1.12	0.75 (1.81)
U4 (high) (%)	0.04	0.63	0.90	2.22	2.18 (3.83)
U4 – U1 (%)	–0.78 (–1.84)	–0.37 (–1.03)	–0.14 (–0.42)	1.18 (2.76)	ANOMALY^{Compustat}_{1997–2014} =1.96*** (5.20)
Panel C. Information uncertainty proxied by 1/COV					
U1 (low) (%)	0.75	0.88	1.03	1.35	0.60 (1.24)
U2 (%)	0.60	0.92	1.04	1.17	0.57 (1.19)
U3 (%)	0.51	0.78	1.05	1.27	0.76 (1.55)
U4 (high) (%)	0.37	0.89	1.10	2.28	1.92 (2.69)
U4 – U1 (%)	–0.38 (–0.72)	0.01 (0.02)	0.07 (0.20)	0.93 (1.98)	ANOMALY^{Compustat}_{1997–2014} =1.31*** (4.02)
Panel D. Information uncertainty proxied by DISP					
U1 (low) (%)	1.06	1.17	1.32	1.69	0.63 (1.36)
U2 (%)	0.69	0.96	1.01	1.20	0.51 (1.11)
U3 (%)	0.42	0.69	0.96	1.26	0.84 (1.42)
U4 (high) (%)	0.11	0.64	0.91	1.88	1.77 (2.12)
U4 – U1 (%)	–0.95 (–1.80)	–0.53 (–1.23)	–0.41 (–1.07)	0.20 (0.52)	ANOMALY^{Compustat}_{1997–2014} =1.14*** (3.68)
Panel E. Information uncertainty proxied by SIGMA					
U1 (low) (%)	0.99	1.05	1.04	1.24	0.25 (0.63)
U2 (%)	0.66	1.00	1.15	1.37	0.71 (1.50)
U3 (%)	0.53	0.86	1.12	1.37	0.84 (1.64)
U4 (high) (%)	0.10	0.56	0.90	1.97	1.87 (2.70)
U4 – U1 (%)	–0.89 (–1.68)	–0.49 (–1.12)	–0.14 (–0.34)	0.74 (1.50)	ANOMALY^{Compustat}_{1997–2014} =1.62*** (4.83)

Table 3. (Continued)

	Momentum quartile				M4 – M1
	M1 (bad news)	M2	M3	M4 (good news)	
Panel F: Information uncertainty proxied by <i>CVOL</i>					
U1 (low) (%)	1.03	1.11	1.12	1.19	0.15 (0.36)
U2 (%)	0.91	1.04	1.07	1.29	0.37 (0.79)
U3 (%)	0.44	0.79	1.13	1.37	0.93 (1.87)
U4 (high) (%)	−0.10	0.52	0.88	1.43	1.53 (2.41)
U4 – U1 (%)	−1.14 (−2.14)	−0.59 (−1.38)	−0.24 (−0.61)	0.24 (0.60)	<i>ANOMALY</i> _{Compustat 1997–2014} = 1.38*** (4.28)

Notes. This table reports the IU-induced underreaction anomaly for all firms covered by the Compustat database during 1997–2014; that is, $ANOMALY_{Compustat 1997–2014}$. The purpose of this analysis is to ensure that the base model and results of Zhang (2006) still hold for our more recent estimation window. We choose 1997–2014 as the estimation window because it is the sample period for our ES-implementation announcements. Columns M1 through M4 report average monthly portfolio returns sorted by price momentum. Rows U1 through U4 report average monthly portfolio returns sorted by IU proxy. In panels A through F, the bottom-right cell reports the estimated value for $ANOMALY_{Compustat 1997–2014}$. A significant and positive number in bold indicates the IU-induced underreaction anomaly. Newey-West *t*-statistics adjusted for autocorrelation are reported in parentheses.

***Significant at the 1% level.

changes in ES implementers' anomaly around the ES go-live (that is, before versus after) relative to a sample of matched control firms. A DiD design increases our confidence that any changes in IU-induced underreaction anomaly are attributable to ES implementation.

Our initial sample includes 748 hand-collected ES implementation announcements. We then exclude firms that experienced other significant business changes during our event window that may also affect market anomaly, which can confound the interpretation of our results. We summarize the procedure in panel A of Table 4 and elaborate in detail below:

i. We exclude 27 firms that substantially increased investment in risky new offerings during the event window. We proxy for such investment by the intensity of exploratory innovation activities. Following the innovation literature (Balsmeier et al. 2017), we calculate a firm's exploratory innovation intensity as the number of exploratory patents filed in a year divided by its total patents filed that year. A patent is classified as exploratory if at least 60% of its citations are based on new knowledge (not in the firm's existing knowledge base). We consider a firm to experience a "substantial" increase in investment in risky new offerings—and exclude it from subsequent analyses—if the firm's annual change in exploratory innovation intensity is in the top 5% among its two-digit standard industrial classification (SIC) industry peers in any year during the event window. Our results remain qualitatively

unchanged if we use the top 1% as an alternative threshold.

ii. We exclude four firms that diversified into foreign markets during the event window. Greater international diversification is associated with greater uncertainty (Duru and Reeb 2002). We use the Compustat segment database to identify firms' changes from a purely domestic firm to an international firm during our event window and exclude them from subsequent analyses.

iii. We exclude 24 firms with substantial changes in the degree of vertical integration during the event window. We measure this by the value-added-to-sales ratio (Scherer and Ross 1990). We consider a firm to have a "substantial" increase in the degree of vertical integration if its annual change in the value-added-to-sales ratio is in the top 5% among its two-digit SIC industry peers in any year during the event window and exclude it from subsequent analyses.

iv. We exclude 25 firms with substantial improvement in information flows to public investors. Following Zhao (2017), we consider a firm to have such improvement if its annual change in 8-K filing frequency is in the top 5% among its two-digit SIC industry peers in any year during the event window and exclude it from subsequent analyses.

After excluding another 96 firms without complete data on firm characteristics for matching, our final sample consists of 572 ES implementers.

Table 4. ES Impact: Propensity Score Matching Diagnostics

Panel A. Sample selection procedure								
Initial ES sample	748							
Firms introducing new risky product lines	(27)							
Firms diversifying into other turbulent markets	(4)							
Firms vertically integrated its suppliers	(24)							
Firms convey clearer messages to its investors	(25)							
Firms with missing matching variables	(96)							
Final ES sample	572							
Panel B. Prematch propensity score regression and postmatch diagnostic regression								
	Prematch dummy = 1 if in ES sample				Postmatch = 0 if in control sample			
<i>Concentration</i>	0.039				0.269			
	(0.097)				(0.270)			
<i>UnrelDiversification</i>	0.323***				0.082			
	(0.054)				(0.142)			
<i>RelDiversification</i>	0.089				−0.651			
	(0.267)				(0.717)			
<i>ROA</i>	0.574***				−0.052			
	(0.091)				(0.230)			
<i>Leverage</i>	0.146**				0.023			
	(0.066)				(0.186)			
<i>Growth</i>	−0.021				0.041			
	(0.026)				(0.077)			
<i>Automate</i>	0.132**				0.076			
	(0.056)				(0.145)			
<i>Transform</i>	−0.063				0.030			
	(0.058)				(0.158)			
<i>HighTech</i>	0.033				0.037			
	(0.045)				(0.115)			
<i>LowTech</i>	0.030				−0.134			
	(0.065)				(0.167)			
<i>1/MV(*1,000)</i>	−7.887***				−1.609			
	(1.155)				(2.452)			
<i>1/AGE</i>	0.193***				−0.076			
	(0.068)				(0.181)			
<i>1/COV</i>	0.005				0.017			
	(0.064)				(0.189)			
<i>DISP</i>	−0.031				4.063			
	(0.425)				(6.318)			
<i>SIGMA</i>	1.918***				−0.426			
	(0.293)				(1.101)			
<i>CVOL</i>	0.182*				−0.119			
	(0.106)				(0.334)			
<i>Constant</i>	−3.826***				−0.403			
	(0.298)				(0.846)			
Year and industry fixed effects	Included				Included			
Observations	200,849				1,144			
Pseudo R^2	0.11				0.02			
P -value of χ^2	<0.000				0.994			
Panel C. Estimated propensity score distributions								
Propensity scores	No. of observations	Mean	SD	Min	P5	P50	P95	Max
ES sample	572	0.007	0.006	0.000	0.001	0.006	0.019	0.053
Control sample	572	0.007	0.006	0.000	0.001	0.006	0.019	0.054
Difference (ES – control)		0.000	0.000	0.000	0.000	0.000	0.000	−0.001

Table 4. (Continued)

Panel D. Differences in firm characteristics between ES sample and control sample				
	(1) ES sample: Post-ES [1, 8] – Pre-ES [–1, –8]	(2) Control sample: Post-ES [1, 8] – Pre-ES [–1, –8]	(3) Difference (ES – Control)	(4) <i>t</i> - statistics
<i>Concentration</i>	0.013	0.015	–0.002	–0.31
<i>UnrelDiversification</i>	0.054	0.072	–0.018	–1.23
<i>RelDiversification</i>	0.022	0.025	–0.003	–0.62
<i>ROA</i>	–0.010	–0.021	0.011	1.36
<i>Leverage</i>	0.014	0.017	–0.003	–0.30
<i>Growth</i>	–0.184	–0.084	–0.100	–1.24
<i>Automate</i>	0.000	0.000	0.000	0.00
<i>Transform</i>	0.000	0.000	0.000	0.00
<i>HighTech</i>	0.000	0.000	0.000	0.00
<i>LowTech</i>	0.000	0.000	0.000	0.00
<i>1/MV</i>	0.001	0.002	–0.001	–1.03
<i>1/AGE</i>	–0.079	–0.073	0.006	0.34
<i>1/COV</i>	0.034	0.024	0.010	0.93
<i>DISP</i>	0.002	0.001	0.001	1.81*
<i>SIGMA</i>	0.001	0.002	–0.001	–0.46
<i>CVOL</i>	0.007	0.013	–0.006	–1.62
<i>ln(Revenue)</i>	0.677	0.644	0.033	0.61
<i>ln(Employee)</i>	0.247	0.290	–0.043	–0.86
<i>IT Investment</i>	0.001	–0.001	0.002	1.37

Notes. This table reports diagnostics for the DiD tests based on PSM for the impact of ES implementation (“ES impact”) on IU-induced underreaction anomaly. Panel A presents the sample selection procedure. Panel B presents parameter estimates from the Probit model used in estimating the propensity scores for the treatment group (ES sample) and control group. The dependent variable is one if the firm-year belongs to the ES sample and zero otherwise. The “Prematch” column contains the parameter estimates of the Probit model estimated using the sample before matching. These estimates are then used to generate the propensity scores for matching. The “Postmatch” column contains the parameter estimates of the Probit model estimated using the subsample of matched treatment-control pairs after matching. *Concentration* is four-firm industry concentration at the four-digit SIC level; *UnrelDiversification* is unrelated diversification measure of the extent to which firms operate across two-digit SIC codes (Dewan et al. 1998); *RelDiversification* is a related diversification measure of the extent to which firms operate across four-digit SIC codes that are within a two-digit SIC code (Dewan et al. 1998); *ROA* is operating income scaled by assets; *Leverage* is total debt divided by total assets; *Growth* is average of the sales growth of the current and previous years; *Automate* is an indicator for whether a client firm belongs to an automate industry, according to Chatterjee et al. (2001) and Anderson et al. (2006); *Transform* is an indicator for whether a client firm belongs to a transform industry, according to Chatterjee et al. (2001) and Anderson et al. (2006); *HighTech* is an indicator for whether a client firm belongs to a high-tech industry, according to Francis and Schipper (1999); *LowTech* is an indicator for whether a client firm belongs to a low-tech industry, according to Francis and Schipper (1999); *1/MV*(*1,000) is one divided by the annual market capitalization (in millions of dollars), multiplied by 1,000; *1/AGE* is the natural logarithm of number of years since the firm was first covered by CRSP by the end of year; *1/COV* is the number of analysts following the firm in the year; *DISP* is the standard deviation of analyst forecasts in the year scaled by the prior year-end stock price; *SIGMA* is the standard deviation of weekly market excess returns over the previous year; *CVOL* is the standard deviation of cash flow from operations in the past five years (with a minimum of three years), where cash flow from operations is earnings before extraordinary items minus total accruals estimated from the balance sheet approach, scaled by average total assets; Year and industry fixed effects are also included. Robust standard errors clustered by firms are displayed in parentheses. Panel C reports the distribution of estimated propensity scores for the ES sample firm-years, control firm-years, and the difference in estimated propensity scores. Panel D reports the differences between ES and control firms’ characteristics in the pre-ES period and post-ES period, respectively. The pre-ES window spans years –8 through –1, with event-year 0 being the year of ES go-live; the post-ES window spans years 1 through 8. SD, standard deviation; P5, 5th percentile; P50, 50th percentile; P95, 95th percentile.

*Significant at the 10% level; **significant at the 5% level; ***significant at the 1% level.

Next, we need to identify a group of control firms that are matched to the treatment firms (ES implementers, in our context). Control firms are also subject to the four exclusion criteria described above. In addition, we acquired a tailored data service from the vendor of the CI database to ensure that our matched control

firms indeed did not implement ES during the event window. Our matching procedure relies on a nearest-neighbor matching of propensity scores (PSM) (Lemmon and Roberts 2010). The major reason for using PSM is to mitigate the endogeneity concern about a firm’s self-choice of ES implementation. The PSM ensures that the

treated firm and the matched control firm have the same propensity of implementing ES before observed endogenous ES implementation.

Operationally, we first run a Probit regression of a dummy that equals 1 if a particular firm-year observation belongs to our ES implementation sample (and 0 otherwise) on a comprehensive list of firm characteristics that could affect a firm's ES implementation decision (Pincus et al. 2017)⁵ and on the six IU proxies. Year and industry dummies are included to capture any time-invariant or industry-specific differences. We then use the predicted probabilities, or propensity scores, from this Probit estimation to perform a nearest-neighbor matching without replacement. That is, we match each ES implementer with the control firm with the closest propensity score.

The Probit model estimates are reported in the first column of panel B of Table 4, labeled "Prematch." They suggest that the specification has substantial explanatory power for the choice variable, as evidenced by a pseudo- R^2 of 11% and a very small p -value for a χ^2 test of the overall model fitness (well below 0.01). The signs of the control variables are generally consistent with prior studies. Unrelated diversification positively affects the propensity to implement ES because ES plays a role in coordinating operations across product segments (Dewan et al. 1998). Firms with more financial resources, as captured by return on assets (ROA) and leverage, are more likely to implement ES (Kobelsky et al. 2008). Most factors related to firms' operating environments (concentration and industry characteristics, except *AUTOMATE*) are insignificant, possibly because we control for industry fixed effects. The coefficient estimate on $1/MV(*1,000)$ is significantly negative, suggesting that smaller firms are less likely to implement ES.⁶ Among the other IU proxies, $1/AGE$, $SIGMA$, and $CVOL$ significantly and positively affect the propensity of ES implementation, which is consistent with prior findings that uncertainty positively affects the likelihood of ES implementation. The coefficient estimates on $1/COV$ and $DISP$ are insignificant, possibly owing to correlation amongst IU proxies. We then use the predicted probabilities, or propensity scores, from this Probit estimation for a nearest-neighbor match without replacement, matching each treatment firm with the control firm with the closest propensity score. This yields 572 treatment firms and 572 control firms.

To ensure matching success, we conduct diagnostic tests. First, we rerun the Probit model restricted to the matched sample and present the Probit estimates in the second column of Table 4, panel B, labeled "Postmatch." No independent variable is statistically significant, suggesting no observable pre-event difference between the two groups of firms. A χ^2 test for overall model fitness shows that we cannot reject

the null hypothesis that all of the coefficient estimates of independent variables are 0 (with a p -value of 0.994).

Second, we check the difference between the propensity scores of treatment firms and of their matched control firms. As reported in panel C of Table 4, the difference is trivial. The maximum difference is only -0.001 ; the median difference is 0.

Our third diagnostic test examines how treatment and control firms change their key time-variant attributes that could also affect their levels of information uncertainty or market anomaly from the pre-ES to the post-ES window. We report the results in panel D of Table 4; graphical illustrations are shown in Appendix B (online). Column (1) presents the average change in firm characteristics before and after ES implementation for ES firms; column (2) presents change in firm characteristics before and after ES implementation for matched control firms. In almost all cases (except *DISP*), there are no significant differences in how ES implementers and nonimplementers change their time-variant attributes from the pre-ES to the post-ES window. In particular, we do not observe a significant difference in change in IT investment between treatment and control firms.⁷ Taken together, the results of these diagnostic tests suggest that our matching process has removed any meaningful observable differences between the ES implementation sample and the control sample.

After obtaining the matched control firms, we compare changes in the IU-induced underreaction anomaly for treatment firms—from an eight-year window $[-8, -1]$ before ES go-live to an eight-year window after ES go-live $[1, 8]$ —to changes in the anomaly for control firms for the same windows.⁸ We exclude the event year in which ES goes live (year 0) to mitigate the effect of potential confounding factors. For treatment firms (ES implementers), we follow the baseline research design (described in Section 3.2) to obtain a 4×4 matrix with different combinations of price momentum quartile and IU quartile and calculate $ANOMALY_{[-8,-1]}^{ES}$ and $ANOMALY_{[1,8]}^{ES}$. We conduct the same calculation for control firms for the pre- and post-ES periods.

Table 5 reports the results. For each panel, the number in the bottom-right cell shows the DiD result, suggesting the impact of ES implementation on the IU-induced underreaction anomaly; a significant and negative number suggests that ES implementation decreases the anomaly. For example, using $1/MV$ as an IU proxy, there was no significant difference in anomaly between ES and control firms during the pre-ES period (0.71% for ES implementers and 0.01% for control firms), but in the post-ES period, ES firms' anomaly became significantly lower (0.85%) than that of control firms (3.21%). The DiD result is -3.06% and significant at the 5% level, suggesting that ES implementers enjoy a 3.06% greater reduction in arbitrage

Table 5. ES Impact: DiD Analysis

	$ANOMALY_{Sample Window}$		
	Window: Pre-ES [−8, −1]	Window: Post-ES [1, 8]	Difference (Post − Pre)
Panel A. Information uncertainty proxied by 1/MV			
Sample: ES (%)	0.71	0.85	0.14 (0.18)
Sample: Control (%)	0.01	3.21	3.20** (2.43)
Diff (ES − control) (%)	0.70 (0.73)	−2.36** (−2.05)	−3.06** (−2.03)
Panel B. Information uncertainty proxied by 1/AGE			
Sample: ES (%)	0.97	0.57	−0.40 (−0.50)
Sample: Control (%)	−0.57	2.82	3.39** (2.55)
Diff (ES − control) (%)	1.54 (1.57)	−2.25** (−2.12)	−3.79*** (−2.62)
Panel C. Information uncertainty proxied by 1/COV			
Sample: ES (%)	0.59	0.12	−0.47 (−0.40)
Sample: Control (%)	0.05	0.27	0.22 (0.22)
Diff (ES − control) (%)	0.54 (0.49)	−0.15 (−0.13)	−0.69 (−0.43)
Panel D. Information uncertainty proxied by DISP			
Sample: ES (%)	−0.21	−0.19	0.02 (0.02)
Sample: Control (%)	−2.94	0.57	3.51** (2.32)
Diff (ES − control) (%)	2.73** (2.48)	−0.76 (−0.64)	−3.49** (−2.18)
Panel E. Information uncertainty proxied by SIGMA			
Sample: ES (%)	0.27	1.59	1.32 (1.07)
Sample: Control (%)	−1.06	1.14	2.20 (1.32)
Diff (ES − control) (%)	1.33 (1.11)	0.45 (0.33)	−0.88 (−0.48)
Panel F. Information uncertainty proxied by CVOL			
Sample: ES (%)	0.70	−0.52	−1.22 (−0.93)
Sample: Control (%)	−1.06	3.16	4.22*** (2.64)
Diff (ES − control) (%)	1.76 (1.32)	−3.68** (−2.29)	−5.44*** (−2.59)

Notes. This table reports DiD analysis for ES impact on the IU-induced underreaction anomaly. Table entries are estimated $ANOMALY_{Sample Window}$. The ES sample consists of firms with ES implementation announcements. For each firm in the ES sample, we identify a control firm using the propensity score matching method. The pre-ES window spans years −8 through −1, with event-year 0 being the year of ES go-live; the post-ES window spans years 1 through 8. In panels A through F, the number in the bottom-right cell indicates ES impact on IU-induced underreaction anomaly; a significant and negative number in bold indicates that ES implementation decreases the IU-induced underreaction anomaly. Newey-West *t*-statistics adjusted for autocorrelation are reported in parentheses.

Significant at the 5% level; *significant at the 1% level.

return (anomaly) after ES implementation than non-implementers. Results based on the other five IU proxies also largely support our hypothesis.

Although collectively the six IU proxies support our hypothesis, they differ in their abilities to detect

the results across different sample periods, sample firms, and empirical designs (as is also the case in our follow-up tests, discussed below). As noted in Section 3.1, a plausible explanation is that the proxies based on the behavior of external stakeholders

Table 6. ES Impact: IT Leaders vs. Non-IT Leaders

	IT leaders: $ANOMALY_{Window}^{Sample}$			Non-IT leaders: $ANOMALY_{Window}^{Sample}$		
	Window: Pre-ES [−8, −1]	Window: Post-ES [1, 8]	Difference (Post − Pre)	Window: Pre-ES [−8, −1]	Window: Post-ES [1, 8]	Difference (Post − Pre)
Panel A. Information uncertainty proxied by $1/MV$						
Sample: ES (%)	0.54	0.88	0.34 (0.20)	Sample: ES (%)	0.27	−0.11 (−0.27)
Sample: Control (%)	−0.36	2.89	3.25* (1.70)	Sample: Control (%)	0.39	1.43 (0.86)
Difference (ES − control) (%)	0.90 (0.45)	−2.01 (−1.12)	−2.91* (−1.64)	Difference (ES − control) (%)	−0.12 (−0.10)	−1.54 (−1.19)
Panel B. Information uncertainty proxied by $1/AGE$						
Sample: ES (%)	0.79	−0.64	−1.43 (−1.20)	Sample: ES (%)	1.94	0.20 (−1.22)
Sample: Control (%)	−1.92	2.54	4.46* (1.72)	Sample: Control (%)	−0.06	1.45 (0.99)
Difference (ES − control) (%)	2.71 (1.23)	−3.18** (−2.02)	−5.89*** (−2.78)	Difference (ES − control) (%)	2.00 (1.01)	−1.25 (−0.74)
Panel C. Information uncertainty proxied by $1/COV$						
Sample: ES (%)	0.23	0.07	−0.16 (−0.10)	Sample: ES (%)	0.82	0.59 (−0.16)
Sample: Control (%)	0.09	0.23	0.14 (0.08)	Sample: Control (%)	−0.05	0.87 (0.62)
Difference (ES − control) (%)	0.14 (0.07)	−0.16 (−0.08)	−0.30 (−0.18)	Difference (ES − control) (%)	0.87 (0.48)	−0.28 (−0.15)
Panel D. Information uncertainty proxied by $DISP$						
Sample: ES (%)	0.13	−0.96	−1.09 (−1.30)	Sample: ES (%)	−0.22	−0.54 (−0.23)
Sample: Control (%)	−1.66	1.30	2.96* (1.69)	Sample: Control (%)	−2.94	0.61 (2.00)
Difference (ES − control) (%)	1.79 (1.25)	−2.26* (−1.77)	−4.05** (−2.11)	Difference (ES − control) (%)	2.72** (1.96)	−1.15 (−0.62)
Panel E. Information uncertainty proxied by $SIGMA$						
Sample: ES (%)	−0.84	1.71	2.55 (1.36)	Sample: ES (%)	0.71	0.11 (−0.35)
Sample: Control (%)	−1.87	1.33	3.31** (1.98)	Sample: Control (%)	−0.24	0.24 (0.21)
Difference (ES − control) (%)	1.03 (0.47)	0.38 (0.28)	−0.76 (−0.23)	Difference (ES − control) (%)	0.95 (0.48)	−0.13 (−0.06)
Panel F. Information uncertainty proxied by $CVOL$						
Sample: ES (%)	0.34	−1.09	−1.43 (−1.01)	Sample: ES (%)	0.92	0.48 (−0.23)
Sample: Control (%)	−0.78	3.76	4.54*** (2.67)	Sample: Control (%)	−1.18	2.37 (2.04)
Difference (ES − control) (%)	1.12 (0.59)	−4.85*** (−2.72)	−5.97*** (−3.01)	Difference (ES − control) (%)	2.10 (1.62)	−1.89 (−1.24)

Notes. This table reports DiD analysis for ES impact on the IU-induced underreaction anomaly for IT leaders and non-IT leaders. Table entries are estimated $ANOMALY_{Window}^{Sample}$. The ES sample consists of firms with ES implementation announcements. For each firm in the ES sample, we identify a control firm using the propensity score matching method. The pre-ES window spans years −8 through −1, with event-year 0 being the year of ES go-live; the post-ES window spans years 1 through 8. A significant and negative DiD result in bold indicates that ES implementation decreases the IU-induced underreaction anomaly. Newey-West *t*-statistics adjusted for autocorrelation are reported in parentheses.

*Significant at the 10% level; **significant at the 5% level; ***significant at the 1% level.

(financial analysts and capital market investors) may be noisier measures of IU than those based on firm characteristics ($1/MV$, $1/AGE$, and $CVOL$). We therefore rely on the overall pattern in the results to draw conclusions. In particular, we observe (a) supporting evidence from both classes of IU proxy—those based on firm characteristics and those based on the

behavior of external stakeholders; (b) stronger evidence from IU proxies based on firm characteristics; and (c) no disconfirming evidence from any IU proxy.

The overall pattern based on these observations supports our hypothesis that ES implementation mitigates the IU-induced market underreaction anomaly.

3.6. Moderation by IT Capability

Enterprise systems are complex, so successful implementation requires sufficient IT capability. Firms with strong IT capability are better placed to realize the benefit of ES in terms of reducing anomaly. To proxy for IT capability (Bharadwaj 2000), we use IT leader ranking by *InformationWeek* (IW) for the period 1992–2008.⁹ We consider an ES implementer to be an IT leader if it appears on the IW ranking at least once in the three years before implementation. We partition our ES sample into two groups according to whether they are IT leaders and repeat the DiD analysis for each. Results are reported in Table 6. We find that IT leaders are better placed than others to attain the anomaly-reducing effect of ES implementation, as reflected by the greater magnitude and the significance levels of the DiD results.

3.7. Complementarity Between Functional and Operational Modules

In this section, we test the complementarity between ES functional and operational modules in mitigating the IU-induced underreaction anomaly. The CI database provides information about implementation of ES modules at the establishment level for each sample firm. Suppose a firm has N establishments in a given year. To measure the extent of implementation of accounting and finance modules, we construct *Acct_Score*, equal to the number of establishments with accounting and finance modules in that year, divided by N . Similarly, we construct *Custm_Score* for customer and sales modules (operational modules supporting downstream customer processes and sales transactions), and *SC_Score*, for supply chain modules (operational modules supporting internal operations and supply chain management processes). Next, we split the CI sample by functional module (accounting and finance) implementation at median into “*Acct_Score*=High” and “*Acct_Score*=Low” and by operational module at median into “*Operation_Score*=High” and “*Operation_Score*=Low,”¹⁰ creating four groups: “HH” (*Operation_Score*=High and *Acct_Score*=High), “HL” (*Operation_Score*=High and *Acct_Score*=Low), “LH” (*Operation_Score*=Low and *Acct_Score*=High), and “LL” (*Operation_Score*=Low and *Acct_Score*=Low).

Table 7 reports results on the complementarity effect between functional and operational modules; entries are the estimated anomalies for the four groups. A significant and positive number (in bold) indicates the IU-induced underreaction anomaly. For the HH group, none of the IU proxies yields a significant anomaly. This is in strong contrast to the significant anomalies for the IU proxies for the HL and LL groups ($1/MV$, $1/AGE$, and $DISP$) and for the LH group ($1/AGE$ and $DISP$). These results suggest that the anomaly-reducing effect of ES is most effective in the HH group, consistent with our

expectation of complementarity between high implementation levels of functional and operational modules.¹¹

4. Supplementary and Robustness Tests

To provide further evidence on the anomaly-reducing effect of ES implementation and to ensure the robustness of our findings, we conduct supplementary and robustness tests. We summarize the findings here; empirical results and more detailed discussion are in the online appendices.

4.1. Mediation Models

As noted, IU stems from two sources: the volatility of a firm’s underlying fundamentals ($\text{var}(v)$) and poor information ($\text{var}(e)$). In Appendix C (online), we test two mediation models on the mechanisms by which ES mitigates the IU-induced underreaction anomaly: (a) ES impact on fundamentals volatility and information quality, which in turn reduces the underreaction anomaly, and (b) module (functional and operational) impact, with functional modules affecting information quality, which in turn reduces the IU-induced underreaction anomaly, and operational modules affecting fundamentals volatility, which in turn reduces the anomaly. We use two proxies for $\text{var}(v)$, *cash flow volatility* and *earnings volatility* (Tian and Xu 2015), and three proxies for $\text{var}(e)$, *management forecast error*, *forecast volatility*, and *forecast specificity* (Dorantes et al. 2013). It is worth noting that all five are public signals to market investors. Investors may not directly observe the inner workings of a firm’s ES and how it changes management and operations in ways that lower $\text{var}(v)$ and $\text{var}(e)$, but if they observe signals conveyed by ES firms that indicate lower $\text{var}(v)$ and $\text{var}(e)$, they will infer that information provided by such firms has lower IU and will therefore react to it in a more timely and complete manner. As shown in Appendix C, we find evidence supporting those two mediation models.

4.2. ES Impact Using Coarsened Exact Matching

Our DiD analyses (Tables 5 and 6) are based on the ES sample and a sample of control firms identified using the propensity score matching method. To ensure that our main findings are not sensitive to the matching algorithm used, in Appendix D (online) we report a robustness test using the nonparametric coarsened exact matching (CEM) method, which results in 432 matched treatment (ES) firms and 432 control firms. We continue to find a significant impact of ES implementation on anomaly reduction—and it is stronger for IT leaders. Note that although PSM and CEM methods and the associated tests help address the endogeneity issue, they may not fully remove it.

4.3. Excluding Twitter Adopters from the ES Sample

In our main DiD analyses, we identified some events that could also affect a firm’s information

Table 7. Complementarity Effect: ES Functional and Operational Modules

ANOMALY ^{Sample} _{2001–2008}			
Sample = HH (Operation_Score=High and Acct_Score=High)	Sample = HL (Operation_Score=High and Acct_Score=Low)	Sample = LH (Operation_Score=Low and Acct_Score=High)	Sample = LL (Operation_Score=Low and Acct_Score=Low)
	Information uncertainty proxied by 1/MV		
–0.06% (–0.08)	1.23%* (1.64)	–0.02% (–0.01)	1.70%* (1.90)
	Information uncertainty proxied by 1/AGE		
0.91% (1.13)	1.39%*** (2.56)	1.07%* (1.66)	0.92%* (1.84)
	Information uncertainty proxied by 1/COV		
0.33% (0.40)	0.56% (1.34)	–0.41% (–0.29)	–0.26% (–0.37)
	Information uncertainty proxied by DISP		
1.36% (1.43)	1.53%** (2.45)	1.44%* (1.72)	2.78%*** (3.32)
	Information uncertainty proxied by SIGMA		
1.18% (1.31)	–1.19% (–1.42)	0.03% (0.01)	0.30% (0.40)
	Information uncertainty proxied by CVOL		
0.93% (0.94)	0.27% (0.46)	0.14% (0.11)	0.11% (0.14)

Notes. This table reports the IU-induced underreaction anomaly for firms covered by the CI database during 2001–2008. We split the sample by two dimensions. First, we split the sample by the extent of functional (accounting and finance) modules implementation at sample median into “Acct_Score=High” and “Acct_Score=Low.” Second, we split the sample by the extent of operational modules implementation; specifically, “Operation_Score=High” means either “Custm_Score=High” or “SC_Score=High”; “Operation_Score=Low” means “Custm_Score=Low” and “SC_Score=Low.” This procedure creates four groups: “HH” (Operation_Score=High and Acct_Score=High), “HL” (Operation_Score=High and Acct_Score=Low), “LH” (Operation_Score=Low and Acct_Score=High), and “LL” (Operation_Score=Low and Acct_Score=Low). Table entries are the estimated ANOMALY in each of the four groups; a significant and positive number in bold indicates the IU-induced underreaction anomaly. Newey–West *t*-statistics adjusted for autocorrelation are reported in parentheses.

*Significant at the 10% level; **significant at the 5% level; ***significant at the 1% level.

uncertainty and excluded from the analyses firms that substantially improved information flow to public investors, proxied by the intensity of change in 8-K frequency (Zhao 2017). The latter phase of our sample period saw explosive growth in the use of social media, enabling managers to release corporate information to public investors. So we use a firm’s launch of Twitter as an alternative measure for enhanced information flows. We acquired a data set from the CI vendor about the year in which firms launched Twitter and repeated the main DiD analyses, excluding 188 firms that launched during the post-ES period. Appendix E (online) shows the results. We continue to find a significant impact of ES on anomaly reduction and a stronger impact for IT leaders.

4.4. Excluding IT Failure Firms from the ES Sample

Our main findings are subject to the concern of failed implementation; we do not know whether ES implementations by our sample firms are successful ex post. We tried to collect announcements of failed ES implementation, but unfortunately—though hardly surprisingly—few firms publicize such failure. What we can find are broad announcements of IT failure. Following prior research (Bharadwaj et al. 2009), we collected such announcements for our 572 ES firms

and repeated the main DiD analyses excluding 22 that made such announcements during the post-ES period. The results are reported in Appendix F (online). We continue to observe a significant impact of ES implementation on anomaly reduction with stronger results for IT leaders.

4.5. Main Analyses Based on 5 × 5 Matrix for Anomaly

Our main tests used a 4 × 4 matrix to calculate the IU-induced underreaction anomaly. To ensure that our results are unaffected by the grouping cutoffs for momentum or IU used to calculate anomaly, we repeated the main analyses based on a 5 × 5 matrix, partitioning observations into IU quintiles and news quintiles. The findings, reported in Appendix G (online), are qualitatively unchanged.

4.6. Analyst Forecast Revisions as Alternate Proxy for News

In the main analyses, we used price momentum—cumulative stock returns over the previous 11 months—as a proxy for news. Another common measure is analyst forecast revisions (used in Zhang 2006). An upward revision, calculated as the average of individual revisions by analysts who covered the firm in

both the current and previous month and scaled by the prior year-end stock price, generally conveys good news about the company, whereas a downward revision signals bad news. We reconduct the complementarity effect test using this alternative measure of news and the corresponding design used by Zhang (2006). We find consistent evidence (Appendix H, online).

4.7. Impact of Individual Modules

In Appendix I (online), we provide finer-grained evidence on the effect of individual ES modules (accounting and finance, customer and sales, and supply chain) in reducing the IU-induced underreaction anomaly. We find largely supportive evidence.

4.8. Shakedown Phase

ES is likely to affect businesses processes, but there is substantial evidence that the firm must first go through a shakedown phase, in which it transitions toward normal operations, before attaining any purported benefits of ES. We examined the time-series pattern of the anomaly-reducing effect of ES implementation. Untabulated results provide some support that the effect is stronger after the shakedown phase (approximately two years in our sample).

5. Discussion

5.1. Implications for Future Research

The foremost implication of our work is that IT implementation by firms can benefit the stock market by reducing market anomaly. Although it has long been argued that aggregating information is one of the market's essential roles (Hayek 1945, Fama 1965), it is often observed that the capital market does not price information in a timely manner, a phenomenon known as market anomaly or market inefficiency. Our findings suggest that a firm's IT can reduce market anomaly—or improve market efficiency—a finding that addresses an essential role of the market and thus has fundamental economic implications. By unveiling IT's business value for the capital market, we contribute to the IT business value literature, in which evidence of IT impact at the market level is sparse. Most studies have examined IT business value at more micro levels, such as firm-level outcomes and worker performance. Our finding at a more macro level (capital market) adds a new dimension of IT impact to that literature and suggests the following research questions.

5.1.1. Which Capital Market Anomalies Can Be Reduced by IT? Research can use dependent variables that tap market anomaly or market inefficiency caused by information uncertainty. IU is widespread, and perhaps for that reason the finance and accounting literature has identified indicators of market anomaly or market

inefficiency attributable to it. These indicators are new dependent variables that can be examined by research on IT business value. For instance, Mendenhall (1991), Abarbanell and Bernard (1992), Ali et al. (1992), and others find that the underreaction of financial analysts—an important group of market participants and professional financial intermediaries—to firm news manifests as a market anomaly. Zhang (2006) finds such underreaction exacerbated by greater information uncertainty. Francis et al. (2007) find investors underreacting to a firm's earnings announcements—a form of market anomaly known as postearnings-announcement drift. Moreover, investors tend to underreact even more in cases of greater information uncertainty. Future research can study the role of IT in reducing such IU-induced market anomalies.

Research can also examine the overreaction anomaly (Bondt and Thaler 1985). Investigating how IT affects the overreaction anomaly will need to be informed by theories that explain it. From a behavioral bias perspective (Daniel et al. 1998), investors tend to overreact to their private information when they are overconfident about their abilities to collect information through private channels (e.g., interviewing managers) and to generate implications from existing information (e.g., corporate financial reports) using proprietary analytics and models. Research can identify types of IT that influence investor activities and evaluate the effect on the overreaction anomaly.

5.1.2. What Economic Value Can IT's Anomaly-Reduction Effect Provide?

Research can consider, at the firm level, an under-researched facet of IT business value—the economic value due to reduced IU. For example, when investors are uncertain about firm information, they require higher return to compensate, raising the cost of equity capital. A logical inference based on our research is that, if IT reduces IU, it can reduce cost of capital and may have implications for cost of debt capital. Bond investors uncertain about a firm's true value will have trouble measuring its real default risk and may require a higher bond premium. These firm-level variables are a facet of IT business value that deserves much more research, given its economic significance.

5.1.3. Which IT Can Reduce the Anomalies and What Other Organizational Resources Complement IT in Doing So?

We examine ES; future research can investigate other IT that mitigates information uncertainty. Our theorizing of an ES impact on IU emphasizes ES impact on a firm's financial information quality and operational variability. Other types of IT can also influence these components, thereby affecting IU. For example, business intelligence systems allow firms to better understand market trends and adjust operations accordingly;

electronic commerce increases geographical scope, dampening sales variation (Dewan and Ren 2011); forecasting software feeds managers accurate information about demand and supply and helps them make informed decisions (Mithas et al. 2013); and interfirm IT allows supply chain partners to work together to sense and respond to market changes. There is also a need to learn more about the interdependencies between technologies and other organizational resources. For instance, how can technologies work in conjunction with other organizational resources to reduce IU? A candidate topic is an internal control system that ensures that managers release firm information faithfully; future work can examine how such a system to regulate managerial action and an information-processing tool jointly affect IU. All in all, examining how firms digitize functional and operational processes by using technologies and combine them with complementary resources to reduce the IU-induced market anomaly is a promising research direction.

5.1.4. In Which Industries Is the Anomaly-Reduction Effect of IT More Likely? The key is to consider where ES is more likely to reduce $\text{var}(v)$ and/or $\text{var}(e)$. Prior research (Brealey and Myers 2002, Lambert et al. 2007) suggests that the volatility of a firm's fundamentals is driven by the nature of its business (e.g., its production system) and industry (e.g., product clock-speed or cyclicity). Firms also differ in the opaqueness of the internal information (Hutton et al. 2009). These prior studies provide theoretical guidance for identifying conditions (e.g., industry sectors with high clock-speed and cyclical dynamics) that better place ES for showing its impact to reduce IU and thus IU-induced anomalies.

5.1.5. What Alternative Mechanisms Allow Investors to Recognize IT's Anomaly-Reducing Effect? Research can take into account the IS literature on how to achieve a successful IT implementation and shorten the shakedown phase for quicker impact on $\text{var}(v)$ and $\text{var}(e)$. However, even with a successful IT implementation, the market relies on certain mechanisms to know that $\text{var}(v)$ and $\text{var}(e)$ have been reduced. Without such mechanisms, the market will not adjust the speed with which it prices the firm's new information. We explored two such mechanisms: management forecasts and actual performance outcome. Examining other information channels is a promising research direction.

5.2. Implications for Practice

Our work has three important practical implications. First, firms should undertake and assess IT implementation using a lens of *reducing transaction frictions with external stakeholders*, including shareholders and

lenders when raising capital. Companies that have strong growth opportunities and pursue innovation need continued external capital support. However, if the cost of capital is high owing to information uncertainty, then capital investment, investment return, and investment in long-term value-enhancing initiatives such as research and development—necessities for long-term success—will all be lower. Implementing IT to reduce information uncertainty can reduce transaction frictions with external stakeholders and thus help the firm raise capital at lower costs.

Second, firms should undertake and assess IT implementation using a lens of *benefits to corporate governance, including effective compensation policies*. Ensuring that security price reflects firm information in a timely and comprehensive manner matters greatly for the governance and compensation policies of publicly traded companies. Stock-price-based metrics are commonly used to better align managers' incentives with firm performance and to mitigate moral hazard. A distortion of security prices due to information uncertainty means that stock prices are no longer a good indicator of the firm's true value and economic reality, so that managers are no longer properly evaluated and compensated. IT can at least partially correct such mispricing.

Finally, our findings have *regulatory implications*. A key role of the market being to aggregate information, high-quality information is a cornerstone of an efficient capital market. Former U.S. Securities and Exchange Commission (SEC) chairman Arthur Levitt (1999) noted that "quality information is the lifeblood of strong, vibrant markets. Without it, investor confidence erodes." Consequently, the SEC has been trying to enhance market transparency and information quality. Our findings suggest that policymakers should encourage public firms to implement IT as part of the overall effort to enhance information quality for investors.

5.3. Limitations and Future Research

Our work has some limitations. First, using ES announcements limits our sample because some firms implementing ES do not make a public announcement. It would be desirable if future research could obtain data directly from ES vendors. Second, our test of complementarity between ES functional and operational modules relies on a statistical method; that is, testing the effect of ES in the copresence of functional and operational modules. Copresence, however, does not necessarily mean that these modules are being used to enhance each other. Future case studies can delve into their actual use. Third, because IU is latent, we have to rely on empirical proxies. Not all proxies are equally effective; for example, stock return volatility (*SIGMA*) seems to be the least reliable

of Zhang's (2006) six proxies. We hope future research can develop new IU proxies so that we can better test IU theory and how IT can mitigate IU-induced anomaly.

6. Conclusion

Our study uncovers an important but unexplored business value of IT: reducing the IU-induced underreaction anomaly in capital markets. Although research has studied impacts of ES implementation on a firm's performance and internal information environment, we surface ES's impact on capital market performance and security pricing, key matters for managers of publicly traded companies and for capital market regulators. We find strong evidence that ES can reduce the ambiguity of new firm information with respect to its implications for firm value, resulting in more timely and complete market reaction to new information and less mispricing of the firm's security. This evidence suggests the important role of IT in reducing market anomaly and improving market efficiency.

Acknowledgments

The authors thank the department editor, associate editor, and three anonymous reviewers for insightful comments. The authors acknowledge the feedback from colloquia presentations at Baruch College, Copenhagen Business School, Georgia Institute of Technology, Hong Kong University of Science and Technology, Iowa State University, Queensland University of Technology, Texas Tech University, University of Dayton, University of Delaware, University of Georgia, University of Groningen, University of Oklahoma, and University of Washington.

Endnotes

¹ Although information available to investors can come from various sources (firm reports, media, analysts), the firm itself is the original source of most of it, because information intermediaries (such as analysts and the media) base their reports and analyses on information released by the firm, formally or informally (Healy and Palepu 2001). Investors' underreaction anomaly can therefore be traced at least partially to the firm's IU.

² Investor overreaction occurs through a different mechanism. According to Daniel et al. (1998), investors tend to overreact to their private information owing to overconfidence. A firm's IT can dampen IU, but reduced IU does not directly affect investors' private information. Research on overreaction needs to address investors' channels/abilities to obtain private information, which drives the overreaction anomaly (Daniel et al. 1998).

³ Other types of IT may also play a role in reducing information uncertainty, to the extent that they can reduce the volatility of firm fundamentals and improve the quality of firm information (Zhang 2006). For instance, electronic commerce increases the geographical scope of a firm's business and so helps dampen sales variation (Dewan and Ren 2011). Forecasting software feeds managers accurate information about demand and supply and helps them make informed decisions (Mithas et al. 2013). Firms can also resort to non-IT means to reduce the volatility of fundamentals. For instance, an efficient product delivery network and capabilities of launching new

products help a firm cope with uncertainties in the market. An effective internal control system reduces the possibility that managers manipulate firm information intentionally (Li et al. 2012).

⁴ Following prior research (e.g., Zhang 2006), we deleted observations for which the absolute value of earnings forecast revision exceeds 100% of the prior year-end stock price because these observations are likely to be erroneous; we also exclude firms with fewer than 12 months of past return data on CRSP to avoid any potential confounding effect of recent initial public offerings.

⁵ Following Pincus et al. (2017), we include the following firm characteristics: *Concentration* is four-firm industry concentration at the four-digit SIC level; *RelDiversification* is a related-diversification measure of the extent to which firms operate across four-digit SIC codes within a two-digit code (Dewan et al. 1998); *UnrelDiversification* is an unrelated-diversification measure of the extent to which firms operate across two-digit SIC codes (Dewan et al. 1998); *ROA* is operating income scaled by assets; *Leverage* is total debt divided by total assets; *Growth* is the average of the sales growth of the current and previous years; *Automate* and *Transform* are indicators for whether a company belongs to an automate or a transform industry, according to Chatterjee et al. (2001) and Anderson et al. (2006); and *HighTech* and *LowTech* are indicators for whether a company belongs to a high-tech or low-tech industry, according to Francis and Schipper (1999).

⁶ MV can capture both firm size and IU. Both large firms and firms with greater uncertainty are more likely to implement ES. In this case, the size effect dominates.

⁷ IT capital investment is calculated following the method of Chwelos et al. (2010).

⁸ It is worth noting that because ES implementers in our sample span the period of 1997–2014, not all have full data for the $[-8, -1]$ and $[1, 8]$ periods. For example, firms that implemented ES in 2011 only have data for the $[-8, -1]$ and $[1, 3]$ periods.

⁹ Since 1991, IW has identified a number of firms each year as the "leaders" of technology in their industries, on the basis of the votes of a group of IT experts. The IT leaders are not ranked on specific IT resources or skills but are peer-ranked on the basis of their overall IT strengths. These peer rankings were used as a measure of a firm's overall IT capability (Bharadwaj 2000). Because IT leaders are not ranked on specific IT resources or skills, it is unlikely that IW ranking was highly intertwined with ES implementation.

¹⁰ "Operation_Score=High" means either "Custm_Score=High" or "SC_Score=High," whereas "Operation_Score=Low" means "Custm_Score=Low" and "SC_Score=Low."

¹¹ We tested alternate partitioning schemes: (a) a 2×2 partitioning based on high versus low accounting and finance modules and customer and sales modules, and (b) a 2×2 partitioning based on high versus low accounting and finance modules and supply chain modules. Results were highly consistent.

References

- Abbaranell JS, Bernard V (1992) Tests of analysts overreaction/underreaction to earnings information as an explanation for anomalous stock price behavior. *J. Finance* 47:1181–1207.
- Ali A, Klein A, Rosenfeld J (1992) Analysts' use of information about permanent and transitory earnings components in forecasting annual EPS. *Accounting Rev.* 67:183–198.
- Anderson MC, Banker RD, Ravindran S (2006) Value implication of investments in information technology. *Management Sci.* 52:1359–1376.
- Ball R, Brown P (1968) An empirical evaluation of accounting income numbers. *J. Accounting Res.* 6:159–178.
- Balsmeier B, Fleming L, Manso G (2017) Independent boards and innovation. *J. Financial Econom.* 123:536–557.

- Barki H, Pinsonneault A (2005) A model of organizational integration, implementation effort, and performance. *Organ. Sci.* 16:165–179.
- Barry CB, Brown SJ (1985) Differential information and security market equilibrium. *J. Financial Quant. Anal.* 20:407–422.
- Bernard VL, Thomas JK (1990) Evidence that stock prices do not fully reflect the implications of current earnings for future earnings. *J. Accounting Econom.* 13:305–340.
- Bharadwaj AS (2000) A resource-based perspective on information technology capability and firm performance: An empirical investigation. *MIS Quart.* 24:169–196.
- Bharadwaj A, Keil M, Mähring M (2009) Effects of information technology failures on the market value of firms. *J. Strategic Inform. Systems* 18:66–79.
- Bhushan R (1989) Firm characteristics and analyst following. *J. Accounting Econom.* 11:255–274.
- Bondt WF, Thaler R (1985) Does the stock market overreact? *J. Finance* 40:793–805.
- Brazel JF, Agoglia CP (2007) An examination of auditor planning judgments in a complex accounting information system environment. *Contemporary Accounting Res.* 24:1059–1083.
- Brazel JF, Dang L (2008) The effect of ERP system implementations on the management of earnings and earnings release dates. *J. Inform. Systems* 22:1–21.
- Brealey RA, Myers SC (2002) *Principles of Corporate Finance* (McGraw-Hill, New York).
- Brown GW (1999) Volatility, sentiment, and noise traders. *Financial Analysts J.* 55:82–90.
- Campbell SD, Sharpe SA (2009) Anchoring bias in consensus forecasts and its effect on market prices. *J. Financial Quantitative Analysis* 44:369–390.
- Chan K, Hameed A (2006) Stock price synchronicity and analyst coverage in emerging markets. *J. Financial Econom.* 80:115–147.
- Chatterjee D, Richardson V, Zmud R (2001) Examining the shareholder wealth effects of announcements of newly created CIO positions. *MIS Quart.* 25:43–70.
- Chwelos P, Ramirez R, Kraemer KL, Melville NP (2010) Does technological progress alter the nature of IT as a production input? New evidence and new results. *Inform. Systems Res.* 21:392–408.
- Cotteleer MJ, Bendoly E (2006) Order lead-time improvement following enterprise information technology implementation: An empirical study. *MIS Quart.* 30:643–660.
- Daniel K, Hirshleifer D, Subrahmanyam A (1998) Investor psychology and security market under- and overreactions. *J. Finance* 53:1839–1885.
- Davenport TH (1998) Putting the enterprise into the enterprise system. *Harvard Bus. Rev.* 76:121–131.
- Dehning B, Richardson VJ, Zmud RW (2007) The financial performance effects of IT-based supply chain management systems in manufacturing firms. *J. Oper. Management* 25:806–824.
- Dewan S, Ren F (2011) Information technology and firm boundaries: Impact on firm risk and return performance. *Inform. Systems Res.* 22:369–388.
- Dewan S, Michael SC, Min CK (1998) Firm characteristics and investments in information technology: Scale and scope effects. *Inform. Systems Res.* 9:219–232.
- Diether KB, Malloy CJ, Scherbina A (2002) Difference of opinion and the cross section of stock returns. *J. Finance* 57:2113–2141.
- Dorantes CA, Li C, Peters GF, Richardson VJ (2013) The effect of enterprise systems implementations on the firm information environment. *Contemporary Accounting Res.* 30:1427–1461.
- Duru A, Reeb DM (2002) International diversification and analysts' forecast accuracy and bias. *Accounting Rev.* 77:415–433.
- Fama E (1965) The behavior of stock market prices. *J. Bus.* 38:34–105.
- Fama E, French KR (1996) Multifactor explanations of asset pricing anomalies. *J. Finance* 51:55–84.
- Forsyth J, Teoh SH, Zhang Y (2007) Misvaluation, CEO equity-based compensation, and corporate governance. Working paper, Pepperdine University, Malibu, CA.
- Foster G, Olsen C, Shevlin T (1984) Earnings releases, anomalies, and the behavior of security returns. *Accounting Rev.* 59:574–603.
- Francis J, Schipper K (1999) Have financial statements lost their relevance? *J. Accounting Res.* 37:319–352.
- Francis J, LaFond R, Olsson P, Schipper K (2007) Information uncertainty and the post-earnings-announcement drift. *J. Bus. Finance Accounting* 34:403–433.
- Freeman RN, Tse S (1989) The multiperiod information content of accounting earnings: Confirmations and contradictions of previous earnings reports. *J. Accounting Res.* 27:49–79.
- Gattiker TF, Goodhue DL (2005) What happens after ERP implementation: Understanding the impact of interdependence and differentiation on plant-level outcomes. *MIS Quart.* 29:559–585.
- Hau H, Lai S (2013) Real effects of stock underpricing. *J. Financial Econom.* 108:392–408.
- Hayek FA (1945) The use of knowledge in society. *Amer. Econom. Rev.* 35:519–530.
- Healy PM, Palepu KG (2001) Information asymmetry, corporate disclosure, and the capital markets: A review of the empirical disclosure literature. *J. Accounting Econom.* 31:405–440.
- Hendricks KB, Singhal VR, Stratman JK (2007) The impact of enterprise systems on corporate performance: A study of ERP, SCM, and CRM system implementations. *J. Oper. Management* 25:65–82.
- Hirshleifer D (2001) Investor psychology and asset pricing. *J. Finance* 56:1533–1597.
- Huber D (2016) The myth of protecting the public interest: The case of missing mandate in federal securities laws. *J. Bus. Securities Law* 16:402–422.
- Hutton AP, Marcus AJ, Tehranian H (2009) Opaque financial reports, R^2 , and crash risk. *J. Financial Econom.* 94:67–86.
- Jegadeesh N, Titman S (1993) Returns to buying winners and selling losers: Implications for stock market efficiency. *J. Finance* 48:65–91.
- Jiang G, Lee C, Zhang Y (2005) Information uncertainty and expected returns. *Rev. Accounting Stud.* 10:185–221.
- Kobelsky KW, Richardson VJ, Smith RE, Zmud RW (2008) Determinants and consequences of firm information technology budgets. *Accounting Rev.* 83:957–995.
- Kohli R, Grover V (2008) Business value of IT: An essay on expanding research directions to keep up with the times. *J. Assoc. Inform. Systems* 9:23–39.
- Kothari SP (2001) Capital markets research in accounting. *J. Accounting Econom.* 31:105–231.
- Lambert R, Leuz C, Verrecchia R (2007) Accounting information, disclosure, and the cost of capital. *J. Accounting Res.* 45:385–420.
- Lang M, Lundholm R (1996) Corporate disclosure policy and analyst behaviour. *Accounting Rev.* 71:467–492.
- Lee HL (2004) The triple-A supply chain. *Harvard Bus. Rev.* 82:102–112.
- Lee HL, Padmanabhan V, Whang S (1997) The bullwhip effect in supply chains. *Sloan Management Rev.* 38:93–102.
- Lemmon M, Roberts M (2010) The response of corporate financing and investment to changes in the supply of credit. *J. Financial Quantitative Analysis* 45:555–587.
- Levitt A (1999) Speech by SEC chairman: Quality information: The lifeblood of our markets. Accessed September 20, 2016, <https://www.sec.gov/news/speech/speecharchive/1999/spch304.htm>.
- Li C, Peters GF, Richardson VJ, Watson MW (2012) The consequences of information technology control weaknesses on management information systems: The case of Sarbanes-Oxley internal control reports. *MIS Quart.* 36:179–203.
- Maibert VA, Soni AK, Venkataramanan MA (2003) The impact of organization size on enterprise resource planning (ERP) implementations in the US manufacturing sector. *OMEGA* 31: 235–246.

- Markus ML, Axline S, Petrie D, Tanis C (2000) Learning from adopters' experiences with ERP: Problems encountered and success achieved. *J. Inform. Technology* 15:245–265.
- Mendenhall R (1991) Evidence of possible underweighting of earnings-related information. *J. Accounting Res.* 29:170–180.
- Mitchell W (1994) The dynamics of evolving markets: The effect of business sales and age on dissolutions and divestitures. *Admin. Sci. Quart.* 39:575–602.
- Mithas S, Tafti A, Mitchell W (2013) How firms competitive environment and digital strategy posture influence digital business strategy. *MIS Quart.* 37:511–536.
- Mukhopadhyay T, Kekre S (2002) Strategic and operational benefits of electronic integration in B2B procurement processes. *Management Sci.* 48:1301–1313.
- Park K, Kusiak A (2005) Enterprise resource planning (ERP) operations support system for maintaining process integration. *Internat. J. Production Res.* 49:3959–3982.
- PeopleSoft (2005) *Customer Relationship Management Customer Success Profiles* (PeopleSoft, Pleasanton, CA).
- Pincus M, Tian F, Wellmeyer P, Xu S (2017) Do clients' enterprise systems affect audit quality and efficiency? *Contemporary Accounting Res.* 34:1975–2021.
- Porter ME, Millar VE (1985) How information gives you competitive advantage. *Harvard Bus. Rev.* 63:149–174.
- Ranganathan C, Brown CV (2006) ERP investments and the market value of firms: Toward an understanding of influential ERP project variables. *Inform. Systems Res.* 17:145–161.
- Reisinger D (2014) Gartner foresees strong IT spending into 2015: 10 key data points. Accessed September 20, 2016, <http://www.eweek.com/it-management/slideshows/gartner-foresees-strong-it-spending-into-2015-10-key-data-points.html>.
- Sambamurthy V, Bharadwaj A, Grover V (2003) Shaping agility through digital options: Reconceptualizing the role of information technology in contemporary firms. *MIS Quart.* 27:237–263.
- Scherer FM, Ross D (1990) *Industrial Market Structure and Economic Performance*, 3rd ed. (Houghton Mifflin, Boston).
- Stratman JK (2001) Information integration for supply chain management: An empirical investigation of ERP systems in manufacturing. Unpublished PhD dissertation, University of North Carolina, Chapel Hill.
- Tian F, Xu SX (2015) How do enterprise resource planning systems affect firm risk? Post-implementation impact. *MIS Quart.* 39:39–60.
- Tversky A, Kahneman D (1974) Judgment under uncertainty: Heuristics and biases. *Science* 185:1124–1131.
- Weill P, Ross JW (2009) *IT Savvy: What Top Executives Must Know to Go from Pain to Gain* (Harvard Business Press, Boston).
- Wixom BH, Todd PA (2005) A theoretical integration of user satisfaction and technology acceptance. *Inform. Systems Res.* 16: 85–102.
- Yao Y, Zhu KX (2012) Do electronic linkages reduce the bullwhip effect? An empirical analysis of the U.S. manufacturing supply chains. *Inform. Systems Res.* 23:1042–1055.
- Zhang XF (2006) Information uncertainty and stock returns. *J. Finance* 61:105–137.
- Zhao XF (2017) Does information intensity matter for stock returns? Evidence from Form 8-K filings. *Management Sci.* 63: 1382–1404.